



32/64-bit End Nodes

32-bit Nodes



Medical
Wearable
Node



Sensor / Motor
Node



Smart Home
Node

Limited Resources
Gateway-assisted PQC

64-bit Nodes



Embedded Linux
Node



Edge-Vision
Node



General-purpose
Node

OS-capable devices
Direct PQC Integration



Local domain AEAD channel
AES-GCM / ChaCha20-Poly1305

Native PQC



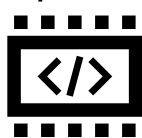
64-bit Gateway Nodes

ML-DSA Module



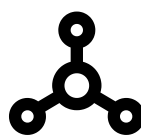
Device
Authentication

SLH-DSA Module
(Optional)



Firmware
Verification

ML-KEM Module



Shared-secret
establishment

KDF

Session-key
derivation

AEAD

Confidentiality
and integrity



Gateway-assisted PQC channel
ML-KEM + KDF + AEAD / ML-DSA / SLH-DSA



Cloud / Server-Side PQC Services



PQC PKI / CA



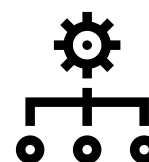
Signed Firmware
Distribution



Secure Data
Storage



Signature
Verification



Key Lifecycle
Management